TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Ultra-high-temperature ceramic marking of technical ceramics for track-and-trace applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Service		Ceramics	
	Medium Category	Small Category	Medium Category	Small Category
	R&D/Engineering service	Product quality control technology	Materials for machines and structures	heat-resistant materials
Concept of Proposed Technology	Due to the very high-temperature processes involved in the production of technical ceramics, it is difficult to establish traceability throughout the entire production chain, which is a key requirement for implementing a smart factory. This is a particular challenge for complex ceramic components in advanced ceramic manufacturing (e.g. alumina, zirconia, silicon nitride or silicon carbide-based materials) that are sintered at temperatures up to about 1,600 °C in harsh kiln atmospheres and subjected to thermal shock. This project aims to address this issue by developing an ultra-high-temperature ceramic marking that can be applied to technical ceramic components using a Drop-on-Demand printing system. The main functions of the technology are (i) to create durable, high-contrast and machine-readable markings that withstand the full ceramic process route (e.g. debinding, high-temperature sintering, glazing/coating, cleaning) and (ii) to enable reliable identification and tracking of individual ceramic parts at each production step. Once this technology has been integrated into production, single-part traceability will be possible, paving the way towards a smart factory in advanced ceramic manufacturing. Potential uses include traceability of high-value technical ceramic components such as kiln furniture, wear- and corrosion-resistant parts, structural ceramics and ceramic components for semiconductor, energy, automotive and aerospace applications, especially where current marking methods fail due to			

Rationale for Joint R&D Project	extreme process conditions. The technology for single-part traceability of technical ceramic components in harsh production environments is strongly related to German core competencies in high-performance ceramics, functional ceramic inks, and industrial printing technology. Technical ceramics are key enabling materials for various Korean strategic industries, including semiconductor and display equipment, secondary batteries, energy systems, and advanced mobility, where reliable traceability and quality assurance are becoming increasingly important. Joint R&D is required because the successful development and commercialization of the technology depend on the combination of both partners' strengths: the research institute contributes advanced know-how in ceramic materials, ink formulation for ultra-high temperatures, and printing/scanning concepts, while Korean SMEs provide access to relevant industrial use cases, ceramic production lines for pilot implementation (e.g. sintering, glazing, machining), and the market knowledge necessary for adapting and scaling the solution to the Korean and wider Asian markets. Only through close collaboration can the technology be validated under real Korean industrial conditions and transferred into commercially viable products and services.
Key Words	1. High-temperature ceramic ink 2. Direct part marking on technical ceramics 3. Track-and-trace / Part traceability 4. Smart manufacturing (Industry 4.0) 5. AI-based process data analytics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	• New ultra-high-temperature ink adapted to specific technical ceramic applications • Scale-up of ink production • Development of a drop-on-demand (DoD) inkjet printer for particle-based (ceramic) inks suitable for technical ceramic substrates • Evaluation of the new printing technology in ceramic production environments (e.g. sintering, glazing/coating lines) • Detection of regular, deformed and/or coated markings on ceramic components to ensure traceability • Development of a data acquisition and (AI-based) analysis framework to link part and process parameters and support smart factory implementation in advanced ceramic manufacturing			
R&D Target Specifications	• Development of a high-temperature ceramic marking for technical ceramic components (temperatures up to 1,500–1,600 °C, depending on application) • Printing technology based on drop-on-demand inkjet printing systems for ceramic particle-based, ultra-high-temperature-stable inks • Reproducible ink production on an industrial scale with consistent quality (batch size > 10 L) • Traceability of ceramic parts throughout the whole production and life cycle (> 95 %) • Improved product quality and reduced production waste • Marking remains machine-readable after completion of the full ceramic process route (e.g. debinding, sintering, glazing/coating, cleaning) by at least 95 % • Reduction of scrap and rework related to misidentified parts by at least 95%			

R&D Plan	The R&D plan is structured into consecutive phases, combining laboratory formulation and testing, pilot-scale validation in ceramic industrial environments, and scale-up of ink production and printing technology. **Pre-study (month 0–6)** • Definition of specifications for technical ceramic applications (temperature range, atmospheres, surface condition) • Test prints with existing high-temperature inks on technical ceramic substrates (green and/or sintered) • Market research on relevant ceramic application fields and potential Korean SME partners • Methodology: literature review, benchmarking of existing marking solutions, preliminary lab tests on ceramic substrates • Milestone: definition of detailed technical specifications and selection of reference ink systems and ceramic use cases **Development of new ink recipe for marking technical ceramics (month 7–19)** • Development of ink with suitable ceramic pigments, binders and additives tailored to technical ceramic substrates and industrial kiln conditions • Test prints on technical ceramic samples in the lab and on pilot production lines (e.g. before/after sintering) • Long-term printing tests including thermal cycling and exposure to typical kiln atmospheres • Methodology: iterative lab-scale formulation, rheology adjustment, adhesion and durability testing under simulated ceramic process conditions • Milestone: formulation of at least one ink candidate fulfilling ultra-high-temperature stability and printability requirements on technical ceramics **Printing technology (month 12–20)** • Modify and test industrial DoD printheads with the new ceramic ink for use on ceramic substrates • Test and vary printing parameters/conditions (e.g. drop size, velocity, stand-off distance, surface preparation) • Detection of markings on technical ceramic samples with optional coatings (e.g. glaze, functional coatings) • Methodology: adaptation and testing of industrial DoD printheads with ceramic particle-based inks, optimization of printing parameters to ensure stable jetting and mark quality on ceramics • Milestone: demonstration of stable printing with required resolution and contrast on relevant technical ceramic components **Ink production (month 15–21)** • Set-up of production equipment (mixers, grinders, etc.) • Scale-up of ink production from lab to industrial scale • Methodology: scale-up of mixing and milling processes, quality control of ink batches (viscosity, particle size distribution, stability) • Milestone: reproducible production of ink batches > 10 L meeting defined quality specifications **Integration of printer and scanner (month 18–26)** • Integration and test of printing technology in ceramic production lines (e.g. before sintering, after sintering, before glazing) • Integration and test of scanner systems for reading markings on ceramic parts in the production line • Testing of track-and-trace functionality in ceramic production environments at Korean SMEs • Milestone: successful end-to-end track-and-trace demonstration in a relevant ceramic industrial environment at a Korean SME **Smart factory (month 24–30)** • Linking of production and part parameters for technical ceramic components • Collection of data in a database for analysis using AI • Data analysis, e.g. improving part quality and reducing waste based on collected data • Milestone: proof-of-concept demonstration of data-driven process optimization (e.g. reduction of scrap rate) in a ceramic pilot line.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Ultra-high-temperature ink development for technical ceramic substrates[1–7] • Concept and support for production line integration of the printer and scanner system in ceramic manufacturing environments • Detection and identification of markings on technical ceramics • Ink technology transfer and training for Korean SME partner(s)
Expected R&R of Korean SME Partner	• Ink production / scale-up for mass production • Integration of printer and scanner equipment into ceramic production lines of end users • Installation of the marking system in the production lines of end users (e.g. manufacturers of technical ceramics, kiln furniture producers, furnace and kiln manufacturers, suppliers of ceramic components for semiconductor/energy/automotive applications) • Data acquisition (aggregation of part numbers and production parameters in a time series), (AI-based) production process / time series analysis for ceramic manufacturing • Provision and coordination of end-user use cases (e.g. technical ceramics for semiconductor, energy, automotive), including access to real production environments for pilot testing

5. Expected Outcomes/Impact

Expected Outcomes and Impact	<u>Expected technical outcomes</u> Development and validation of a novel, ultra-high-temperature and chemically resistant ceramic ink system for direct marking of technical ceramic components. The ink will withstand process temperatures up to 1,500–1,600 °C and harsh kiln/process environments (e.g. high-temperature sintering, reactive atmospheres, glazing/coating, cleaning) without losing legibility. The project aims to increase the Technology Readiness Level from TRL 3 to TRL 7 by demonstrating stable performance in relevant industrial ceramic pilot lines at Korean SMEs. <u>Impact on manufacturing, quality and traceability</u> The new ink will enable robust, part-level traceability throughout the entire ceramic production chain, including ultra-high-temperature and chemically aggressive process steps where conventional markings fail. By reliably linking each individual ceramic part to its specific process and production parameters, manufacturers can: • Reduce misidentification and mix-ups of parts by more than 95 %. • Enable systematic root-cause analysis and continuous process optimization based on real production data. • Improve outgoing quality and reduce scrap, rework, and recall risks in technical ceramic production. <u>Economic and commercial impact</u> For Korean SMEs in the technical ceramics value chain, the technology will lower costs related to manual labeling, rework, scrap, and quality incidents, while enabling higher process transparency and more efficient quality assurance. The ink system and associated marking process can be commercialized as: • A new high-performance ceramic ink product line for ultra-high-temperature ceramic applications. • A complete marking solution (ink + application guidelines, integration with printer and scanner systems) for technical ceramic manufacturers and related industries (e.g. kiln furniture, ceramic components for semiconductor, energy, automotive and aerospace). Within 2–3 years after project completion, the partners expect initial sales in domestic markets and subsequent expansion into international niche markets requiring high-temperature traceability for

	technical ceramics. Impact for Korean SMEs and industrial ecosystem The project will equip Korean SMEs with a key enabling technology for digital and data-driven manufacturing in the field of technical ceramics. By making each ceramic component traceable across all process steps, SMEs can implement robust track-and-trace systems and advanced quality analytics, which are essential elements of smart manufacturing and Industry 4.0. This strengthens the competitiveness of Korean suppliers in global value chains and supports the development of a local ecosystem for high-value, traceability-enabled technical ceramics. Scientific and technological impact Scientifically, the project will advance understanding of ultra-high-temperature-stable pigments and binders for ceramic-on-ceramic marking, ink–ceramic surface interactions under extreme thermal and chemical conditions, and the design of codes/marks that remain machine-readable after full ceramic process cycles. The results will form a basis for future R&D on functional ceramic inks and coatings for harsh environments and will be disseminated through joint publications, conference presentations, and, where appropriate, patent applications. Environmental and societal impact By reducing scrap rates, rework, and product recalls through improved traceability and process control in technical ceramic production, the technology contributes to more resource- and energy-efficient manufacturing. In addition, reliable automated marking and identification can reduce manual handling and checking of hot or fragile ceramic parts, thereby improving occupational safety and lowering the physical burden on workers in Korean SMEs. Overall, the joint R&D project will provide Korean SMEs with a robust key technology for high-temperature part traceability and smart manufacturing in the field of technical ceramics, while enabling the research institute to transfer and further develop its expertise in high-performance ceramic inks and industrial printing solutions in collaboration with strong industrial partners.

6. References

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